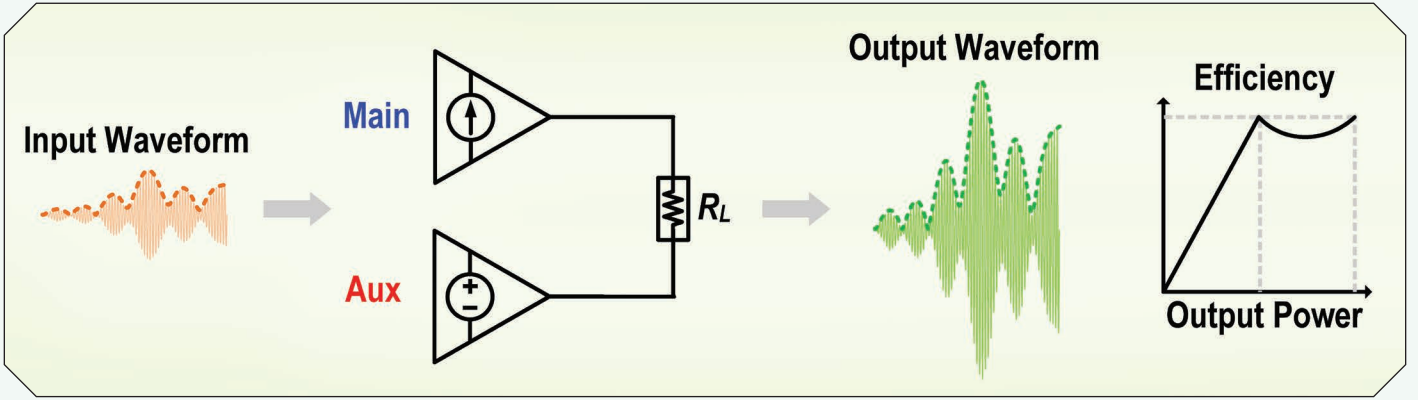


Doherty Power Amplifiers

Fundamentals and recent advances



THE OPENER ART WAS DRAWN BY THE AUTHOR (QIANG ZHOU)

Introduction

The PA is the most power-hungry block in a wireless transceiver front end, making energy efficiency a key design target. However, achieving high PA efficiency in modern wireless communication systems is challenging because of the widespread adoption of high-order QAM, OFDM, and carrier aggregation. While these techniques improve link throughput and robustness, they also introduce a high PAPR, as high as 10–12 dB for 5G New Radio (NR) and Wi-Fi waveforms. As a result, PAs spend most of their time in the deep-back-off region, where conventional linear-mode PAs are inherently inefficient.

This challenge is further illustrated in Figure 1. Although the theoretical peak efficiency of an ideal class-B PA reaches 78.5%, the efficiency at 12-dB back-off drops to only one quarter of this value. When weighted by the probability density function (PDF) of the signal, the average efficiency is only ~20%. For communication PAs, back-off efficiency and average efficiency are therefore far more relevant metrics than peak efficiency.

The Doherty PA, invented by William Doherty at Bell Laboratories in 1936 [1], provides an elegant solution to realize high back-off efficiency. Despite being 90 years old, it remains a dominant topology for cellular infrastructure, with recent CMOS implementations extending its reach from RF to mmWave. This article reviews the fundamentals of the Doherty PA, its architectural variations, the two key design ingredients for successful implementation, and recent advances in bandwidth exten-

sion, multiway deep-back-off efficiency enhancement, and integration into mmWave beamformers.

Fundamentals of Doherty Active Load Modulation

Efficiency Limitation of Class-B PA

To appreciate the ingenuity of the Doherty PA, it is essential to first understand the efficiency limitation of a conventional linear-mode PA. In an ideal class-B PA (see Figure 2), the transistor gate is biased at its threshold voltage, resulting in a half-sinusoidal waveform for the drain-to-source current I_{ds} . The drain efficiency (η) is calculated as the ratio of the fundamental RF output power to the DC power, as

$$\eta = \frac{1}{2} \cdot \frac{V_{ds}(f_0)}{V_{DD}} \cdot \frac{I_{ds}(f_0)}{I_{DC}} \quad (1)$$

where $V_{ds}(f_0)$ and $I_{ds}(f_0)$ are the fundamental amplitudes of V_{ds} and I_{ds} , respectively.

At peak power [see Figure 2(a)], I_{ds} reaches its peak value, and the drain voltage V_{ds} also reaches its full swing from 0 to $2V_{DD}$. When the input power is backed off [see Figure 2(b)], the magnitude of I_{ds} decreases but remains half sinusoidal, keeping the ratio of $I_{ds}(f_0)/I_{DC}$ constant. However, since the load impedance R_L is fixed, the fundamental drain voltage $V_{ds}(f_0) = I_{ds}(f_0) \cdot R_L$ decreases in proportion to the reduced $I_{ds}(f_0)$. Referring back to (1), since $V_{ds}(f_0)/V_{DD}$ decreases while the current ratio remains constant, the efficiency η drops linearly with $V_{ds}(f_0)$. In other words, the reduction in drain voltage swing is the root cause of back-off efficiency degradation.

This analysis leads to a critical insight: imagine an ideal load tuner that increases R_L as the input power decreases and compensates for the reduced current $I_{ds}(f_0)$ [see Figure 2(c)], then $V_{ds}(f_0)$ would maintain its maximum swing of V_{DD} regardless of the power level, thereby achieving consistently high efficiency throughout back-off. This is the key idea of load modulation: *increasing the load impedance presented to the transistor as the input power decreases*.

Original Doherty Invention and Series Versus Parallel Doherty Architectures

In William Doherty's original invention [see Figure 3(a)], the circuit consists of a main amplifier (modeled as a current source), a "hypothetical" auxiliary amplifier (modeled as a voltage source), and a series load resistor $2R$. At peak power, both amplifiers turn on and deliver equal power to the load, meaning that each amplifier sees an equivalent resistance of R . In the small-signal region, the auxiliary voltage source turns off and becomes a short circuit, which doubles the impedance seen by the main amplifier to $2R$. This aligns perfectly with the goal of load modulation: presenting a higher impedance to the main amplifier at lower power levels.

Another brilliant idea in Doherty's invention was recognizing that a current source paired with an impedance

inverter is electrically equivalent to a voltage source [see Figure 3(b)]. When the auxiliary current source is off (i.e., acting as an open circuit), the impedance inverter transforms this open circuit into a short circuit, recreating the required voltage source behavior.

Redrawing Doherty's original invention using modern circuit symbols and implementing the impedance inverter using a $\lambda/4$ transmission line (Tline) yields the schematic in Figure 3(d). Since the series load R_L can be inconvenient for single-ended antennas, a series-combining transformer is inserted to isolate the load from the amplifiers. At peak power, both the main and auxiliary amplifiers are on, and each sees an impedance of $0.5R_L$.

When the auxiliary amplifier is off, the impedance inverter transforms its open circuit to a short circuit, shorting the lower transformer to ground. As a result, the main amplifier sees an impedance of R_L , which is twice its peak-power impedance. This architecture is often referred to as the "series Doherty" architecture because the main and auxiliary amplifiers are connected in series.

Doherty also realized that the impedance inverter could instead be placed at the main amplifier's output, creating the "parallel Doherty" architecture [see Figure 3(c)]. Redrawing using modern circuit symbols (where the characteristic impedance of the $\lambda/4$ line is $2R_L$), both amplifiers see an impedance of $2R_L$ at peak power

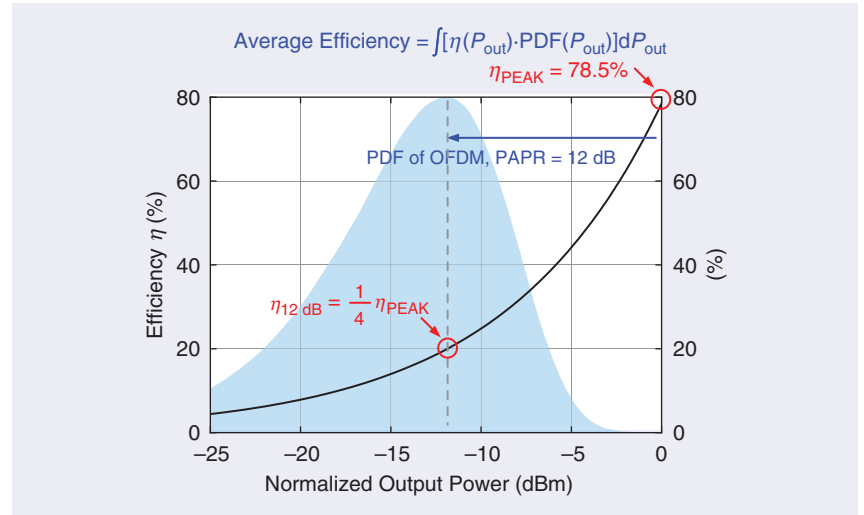


FIGURE 1: Efficiency versus back-off for an ideal class-B PA, overlaid with the PDF of an OFDM signal having a 12-dB PAPR, highlighting the mismatch between peak efficiency and back-off efficiency. PDF: probability density function.

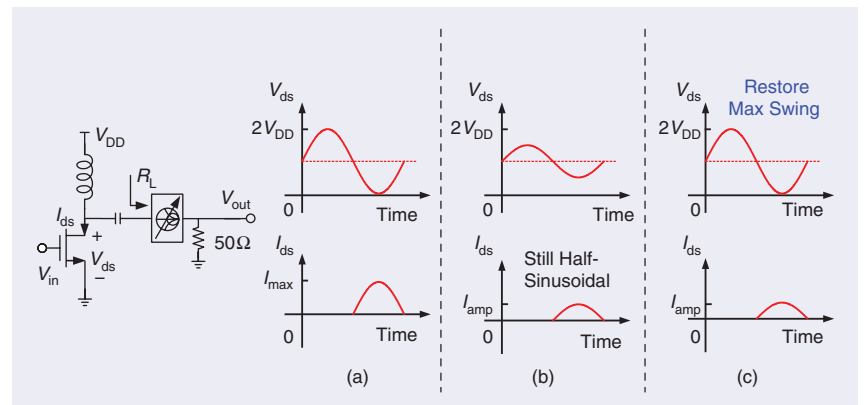


FIGURE 2: Drain voltage (V_{ds}) and drain-to-source current (I_{ds}) waveforms for an ideal class-B PA at (a) peak power, (b) back-off, and (c) back-off with an ideal load tuner that increases R_L as the input power decreases.

[see Figure 3(e)]. When the auxiliary amplifier is off, the impedance inverter transforms the load impedance R_L into $4R_L$ for the main amplifier. Once again, the impedance seen by the main amplifier is doubled when the auxiliary amplifier is off. The parallel Doherty architecture is generally more widely adopted than the series Doherty architecture as current combining is easier to implement than voltage combining.

In addition to the two basic Doherty PA architectures, several useful variations, employing different series/parallel combiner topologies and current/voltage-source auxiliary amplifiers, are summarized in [2].

Both series and parallel Doherty architectures exhibit two theoretical efficiency peaks: one at peak power when both amplifiers achieve maximum voltage swing, and another when the main amplifier reaches full swing just before the auxiliary amplifier turns on. In practice, the second efficiency peak is typically lower than the first because of several nonidealities: the transistor draws a nonzero quiescent current (which is neglected in the ideal analysis), the loss of the output passive network varies across power levels, and the impedance presented to the main amplifier may fall short of the desired $2\times$ increase when the auxiliary amplifier is off [3].

Key Ingredients: Output Network Synthesis and Auxiliary Control

Achieving first-pass design success for a Doherty PA relies on two key design ingredients: 1) the Doherty output network, which must realize the desired active load modulation in a low-loss, compact form factor, and 2) cooperation between the main and auxiliary amplifiers, ensuring that the auxiliary turns on at the correct power level and ramps up its current with the desired slope. This section delves into practical design considerations for these two key ingredients.

Doherty Output Network Synthesis

Impedance Inverter Based on Lumped Elements

A critical component in the Doherty output network is the impedance inverter. The most straightforward implementation is a $\lambda/4$ Tline. While widely used in PCB and III-V MMIC designs, where wide, low-loss Tlines are readily available, this approach is prohibitively bulky and lossy for CMOS PAs, especially at lower frequencies.

To reduce size, designers often approximate the $\lambda/4$ line using a first-order lumped CLC π -network [see Figure 4(a)]. The shunt capacitors include both device parasitic capacitances and any additional explicit capacitors. Two early works [4], [5] adopted this

approach and, for the first time, demonstrated the feasibility of designing Doherty PAs in CMOS.

A few other variations of the CLC π -network are shown in Figure 4. In particular, an on-chip transformer, along with two shunt capacitors, can realize an impedance inverter when sized correctly [6]. The network synthesis starts with the T model of the transformer, followed by a T-to- π conversion. The shunt inductor is then split into two parallel branches: one branch provides the negative L required by the π network, and the other branch is resonated out by a shunt capacitor. Because a transformer inherently merges two inductors into a single magnetic footprint, this approach is highly area efficient.

Since CMOS PAs typically require impedance down-transformation from the 50- Ω load, the impedance inverter is often jointly designed with the impedance down-transformation to minimize overall size and loss.

Impedance Inverter Based on Distributed Elements at mmWave

Although transformers are powerful workhorses at lower frequencies, their lumped-element models exhibit noticeable mismatches with electromagnetic (EM) simulations as the frequency increases, as a result of various nonidealities. On the other hand, distributed elements, such as coupled lines

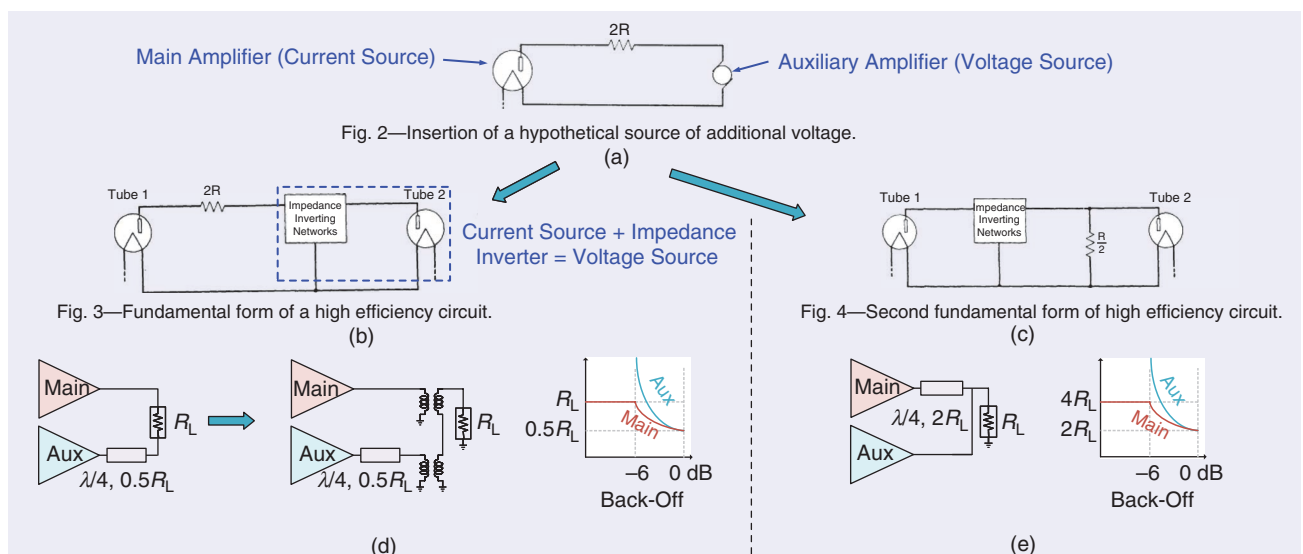


FIGURE 3: (a)–(c) Doherty's original invention and two classical Doherty architectures: the series Doherty and the parallel Doherty, with captions from [1]. (d) and (e) We redraw the (d) series and (e) parallel Doherty architectures using modern circuit symbols and their load modulation.

and couplers, become affordable on chip at higher frequencies. Their behavior is fully determined by the characteristic impedance and electrical length, and the resulting model aligns closely with EM simulations. This makes them attractive candidates for Doherty network implementation at mmWave.

To systematically determine the design parameters of a distributed-element-based Doherty network, we turn to network theory. For a parallel Doherty network, we derive its three-port Z-matrix by sequentially driving each port with a current source while open-circuiting the remaining ports [8]. The resulting Z-matrix is shown in Figure 5(a). Note that the corresponding Y-matrix does not exist since the determinant of the Z-matrix is zero. Intuitively, deriving the Y-matrix requires driving one port with a voltage source while short-circuiting the others to ground, which is not physically feasible when the Aux and Out ports are already connected. By duality, a series Doherty network is described by a three-port Y-matrix, but not a Z-matrix, as shown in Figure 5(b). Once these canonical matrices are derived, the key insight is that any physical network whose Z- or Y-matrix matches one of these forms is guaranteed to realize the correct Doherty load modulation.

Figure 5(c) and (d) illustrates this principle using a quadrature hybrid [7]. A quadrature hybrid has four ports: three of them are connected to the main amplifier, the auxiliary amplifier, and the load, while the fourth port needs to be terminated. If the fourth port is terminated by an open circuit, the resulting three-port Z-matrix matches the parallel Doherty form [see Figure 5(c)]. Similarly, if the fourth port is terminated by a short circuit, the resulting three-port Y-matrix matches the series Doherty load form.

Another example employs an extended rat-race coupler [8], whose Z-matrix is equivalent to two sets of parallel Doherty PAs combined in series. This multifunction integration is particularly powerful for simultaneously achieving back-off efficiency enhancement and watt-level output power at mmWave.

Simulating Doherty Load Modulation Behavior

Before co-simulating the Doherty output network with active transistors in practical designs, it is critical to first

validate the network's load modulation behavior using electronic design automation software. An example testbench is available at [9]. In this setup, the main and auxiliary amplifiers are

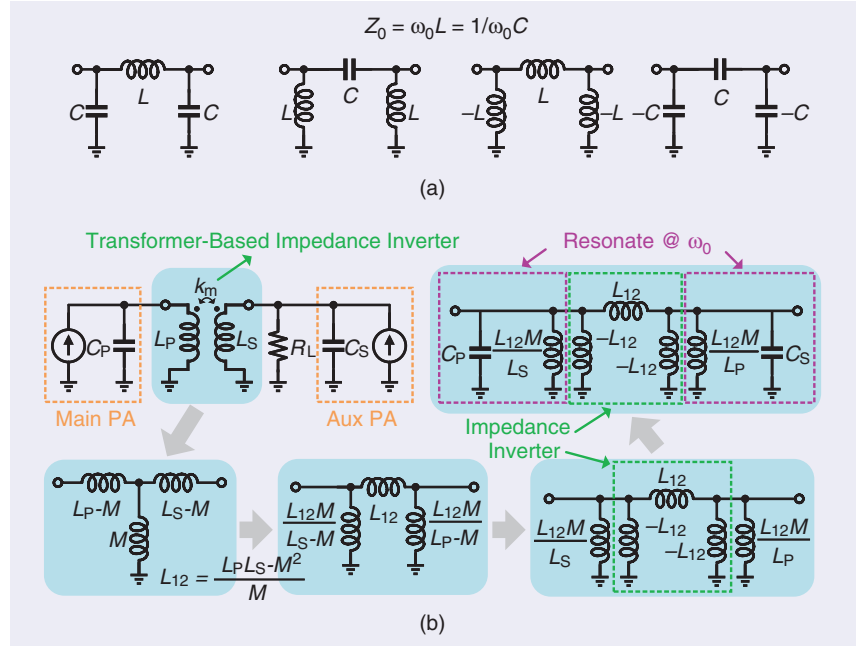


FIGURE 4: Impedance inverters using lumped elements: (a) π -networks and (b) transformer-based impedance inverter synthesis.

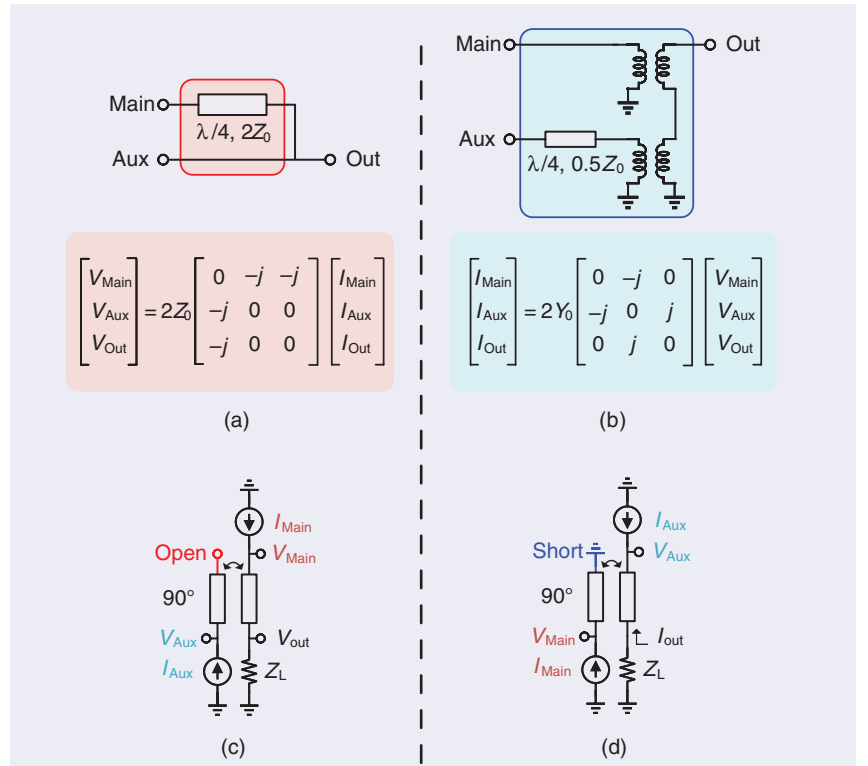


FIGURE 5: Three-port (a) Z-matrix of the parallel Doherty network and (b) Y-matrix of the series Doherty network. A quadrature hybrid as a (c) parallel or (d) series Doherty network by terminating the fourth port with an open or a short circuit [7].

modeled as ideal current sources with a 90° phase offset. Current probes are used to monitor the impedance presented to each amplifier as the input voltage increases. The current profile is consistent with the classical parallel Doherty theory: the main amplifier's current ramps up linearly from zero to its peak, while the auxiliary amplifier's current remains zero until 6-dB back-off and then increases linearly.

Two simulation outputs are critical for design validation: 1) the real and imaginary impedances presented to each path across power levels, which should match the theoretical load modulation curves in Figure 3, and 2) the passive loss across power levels. These results provide direct, quantitative feedback on the network's performance and serve as guidance for further tuning and optimization. Note that transistor parasitic capacitances should be included as part of the network in these simulations [10].

Cooperation Between Main and Auxiliary Amplifiers

The second key ingredient is the precise control of the auxiliary amplifier.

The simplest way is to bias the auxiliary amplifier in class C, so that it remains off in the low-power region. However, a class-C-biased auxiliary of the same transistor size as the class-AB main amplifier produces a smaller peak current. In other words, while class-C biasing controls when the auxiliary turns on, it does not provide effective control over the current slope.

Analog Auxiliary Control Through Adaptive Biasing

Adaptive biasing overcomes the limitations of fixed class-C biasing. It typically consists of two stages: an envelope detector that extracts the AM of the RF input, followed by a buffer that converts this envelope to a dynamic bias for the auxiliary amplifier [see Figure 6(a)]. As the input power increases, the bias ramps from class C to class AB, providing control over both the turn-on point and the current slope. A design example with high programmability and PVT robustness can be found in [11].

Bandwidth is a key bottleneck of the adaptive biasing circuit. Modern wireless standards impose demanding bandwidth requirements: 5G FR2

user equipment requires >100-MHz modulation bandwidth, the infrastructure requires >2000 MHz, and satellite broadband systems have similarly stringent requirements. Since the AM bandwidth is approximately three times the modulation bandwidth (e.g., a 400-MHz modulated signal requires a 1.2-GHz AM tracking bandwidth), it can be challenging for the buffer stage to drive the large parasitic capacitance of the PA power cell at high speeds.

A simple yet effective method to increase the bandwidth is to add a shunt-peaking inductor (L_{S2}) into the buffer [12], as shown in Figure 6(b). With an optimal L_{S2} of 700 pH, the bandwidth is extended by 87% without introducing overshoot [see Figure 6(c)]. Transient simulations also confirm that the envelope-tracking error remains small up to a 2000-MHz modulation bandwidth. It is worth noting that the quality factor of L_{S2} is not a concern since a series resistor R_S is inherently needed to provide sufficient buffer gain. Multiple metal layers can therefore be stacked to implement this shunt-peaking inductor with minimal area overhead.

Digital Auxiliary Control Through Digital PA

An alternative to adaptive biasing is to control the auxiliary amplifier using digital power cells, an idea inspired by the digital PA architecture. The auxiliary turn-on point and the cooperation between the main and auxiliary are set by digital codewords that determine how many unit transistor cells are turned on. The first digital Doherty PA was demonstrated in [13] using a parallel Doherty architecture [see Figure 7(a)]. Subsequently, digital series Doherty PAs were reported [14]. A switched-capacitor PA naturally behaves as a voltage source in series with a capacitor [15], making it an ideal fit for the auxiliary voltage source required in the series Doherty architecture.

Recent Advances

Despite its clear efficiency advantage, Doherty PAs face several practical limitations. 1) The $\lambda/4$ Tline, or its equivalents, behaves as an ideal

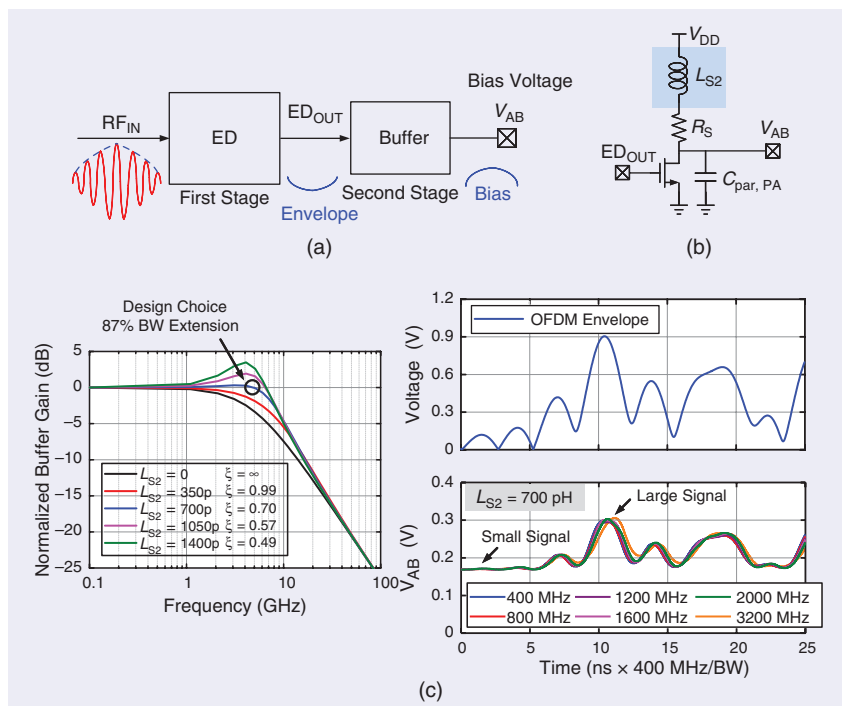


FIGURE 6: (a) Adaptive biasing circuit diagram. (b) Adding a shunt-peaking inductor L_{S2} to increase the buffer bandwidth. (c) Simulated buffer bandwidth versus L_{S2} and transient waveforms of the gate bias voltage V_{AB} under 400–3200-MHz modulation bandwidths [12]. ED: envelope detector.

impedance inverter only at the center frequency, resulting in limited bandwidth. 2) A conventional two-way Doherty PA provides efficiency enhancement only up to 6-dB back-off, which is insufficient for OFDM signals with a 10–12-dB PAPR. 3) Doherty PAs incorporate a main path and an auxiliary path, which doubles the area compared to a single-path PA. 4) Doherty PAs often require linearization, such as digital predistortion (DPD); while the power consumption of DPD is manageable for high-power base-station PAs, it becomes appreciable for silicon PAs. 5) Doherty PAs can be sensitive to antenna VSWR variations, degrading the efficiency gain when the load deviates from 50 Ω . These limitations have historically restricted Doherty PAs to cellular infrastructure, while average power tracking and envelope tracking have dominated handset PA design [16]. This section presents recent advances that address several of these challenges, with a focus on bandwidth extension, deep-back-off efficiency enhancement, and size reduction.

Overcoming the RF Bandwidth Bottleneck

In a parallel Doherty PA, the bandwidth bottleneck is the $\lambda/4$ impedance inverter (TL_1) in the main path (see Figure 8). At peak power, TL_1 provides a 1:1 impedance transformation. However, at 6-dB back-off when the auxiliary amplifier is off, TL_1 transforms $0.5Z_{opt}$ to $2Z_{opt}$, resulting in a large 1:4 impedance transformation ratio that limits the bandwidth. A more detailed analysis of the frequency response of a parallel Doherty PA is presented in [17].

Given that the 1:4 transformation ratio of TL_1 is the bottleneck, how can we extend the bandwidth?

Series Doherty

The first technique is to remove TL_1 and adopt the series Doherty architecture. In this case, the main and auxiliary amplifiers are connected in series without the $\lambda/4$ impedance inverter, which inherently supports a wider RF bandwidth. The digital series Doherty

PA in [14] confirms this advantage, demonstrating consistently high efficiency and strong back-off improvement across 750–1050 MHz.

Three-Tline Doherty

The second technique is to remove TL_2 (i.e., the Tline between the combining node and the antenna), so that TL_1 is directly loaded by the antenna impedance Z_{ANT} instead of the $0.5Z_{opt}$ at back-off. This changes the impedance transformation ratio of TL_1 from 1:4 to $Z_{ANT}:2Z_{opt}$, which can be substantially smaller for typical Z_{opt} values. To maintain correct phase and impedance looking into the auxiliary path, two additional Tlines are added to the auxiliary amplifier's output, resulting in the three-Tline Doherty architecture [see Figure 9(a)].

Figure 9(b) plots the impedance presented to the main amplifier across frequency for a conventional parallel Doherty and the new three-Tline Doherty [6], [18], [19]. A substantially flatter impedance profile is achieved for the three-Tline Doherty, demonstrating its bandwidth advantage.

Despite requiring three Tlines conceptually, the entire network can be realized in a two-transformer footprint using the lumped-element impedance inverters presented in the section “Impedance Inverter Based on Lumped Elements,” keeping the area comparable to that of the conventional parallel Doherty counterpart.

Multiway Doherty for Deep-Back-Off Efficiency Enhancement

A conventional two-way Doherty PA provides efficiency enhancement only up to 6-dB back-off, which is insufficient for high-PAPR signals, such as 5G NR OFDM. In theory, an N -way Doherty PA introduces N efficiency peaks, extending the efficiency enhancement to $20 \log(N)$ dB of back-off when all N amplifiers are equally sized. Figure 10 compares the achievable efficiencies of two-, three-, and four-way Doherty PAs [12]. While the four-way Doherty PA exhibits a lower peak efficiency from the additional 90° Tlines in its output network, it achieves a $1.44 \times$ power savings compared with the two-way Doherty PA.

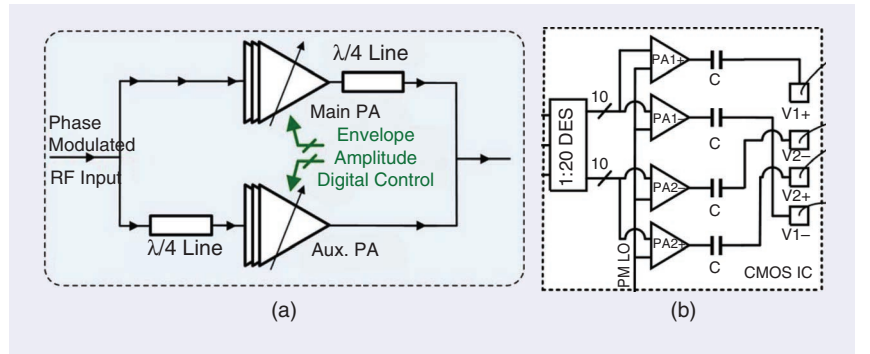


FIGURE 7: (a) Digital parallel Doherty [13]. (b) Digital series Doherty [14].

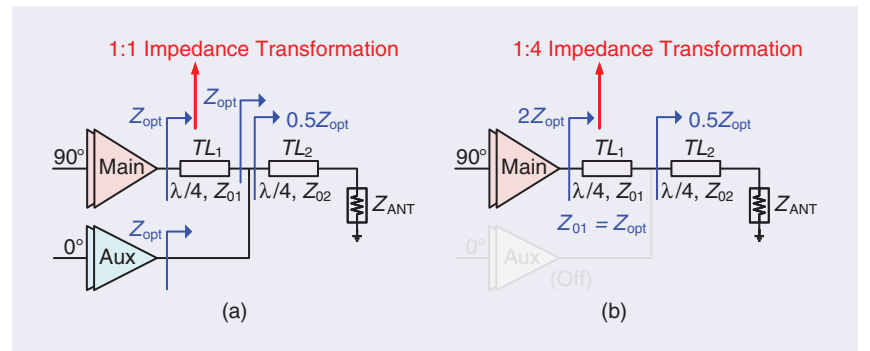


FIGURE 8: Impedance transformation of TL_1 at (a) peak power and (b) 6-dB back-off.

Before delving into the multiway Doherty, we first highlight a key insight into the $\lambda/4$ Tline in the two-way

Doherty PA. The voltage (or current) at one end of the Tline is uniquely determined by the current (or voltage) at

the other end, as shown in Figure 11. In a two-way parallel Doherty PA, the load current I_L increases linearly from zero to its peak value, while the auxiliary amplifier's output current I_A turns on only in the high-power region. As a result, the current at the output of the impedance inverter, $I_0 = I_L - I_A$, remains constant between 6-dB back-off and peak power. Through the impedance inverter, the main amplifier's output voltage V_M follows I_0 , which also remains constant in the high-power region, resulting in high efficiency.

Moving on to the three-way Doherty, two architectures have been reported in the literature. The first is a "cascaded" three-way Doherty [see Figure 12(a)]: the main amplifier and auxiliary amplifier #1 form a two-way parallel Doherty, which is then combined with auxiliary amplifier #2, again in a parallel Doherty fashion. Unfortunately, this architecture has a critical flaw. To realize the first efficiency peak at 9.5-dB back-off, the main amplifier's output voltage V_M must saturate at this power level. Similarly, to realize the second efficiency peak at 6-dB back-off, the output voltage of auxiliary amplifier #1 (V_{A1}) must also saturate at 6-dB back-off. Since the main amplifier's output current I_M follows V_{A1} through the impedance inverter, both V_M and I_M are saturated at 6-dB back-off. This means that the main amplifier no longer outputs additional power beyond this point, raising significant linearity and reliability concerns. If early saturation is avoided by allowing I_M to ramp up linearly to peak power, then V_{A1} follows this linear trajectory through the impedance inverter between V_{A1} and V_M , reaching its peak value only at peak power. This fails to achieve the desired V_{A1} profile and the resulting efficiency peak at 6-dB back-off.

The correct three-way Doherty is shown in Figure 12(b). Here, a parallel two-way sub-Doherty is combined with the main amplifier through a series combiner [20]. Its operating principle is explained as follows [3]. The main amplifier's output current I_M ramps up linearly to peak power, ensuring that the main amplifier is inherently free

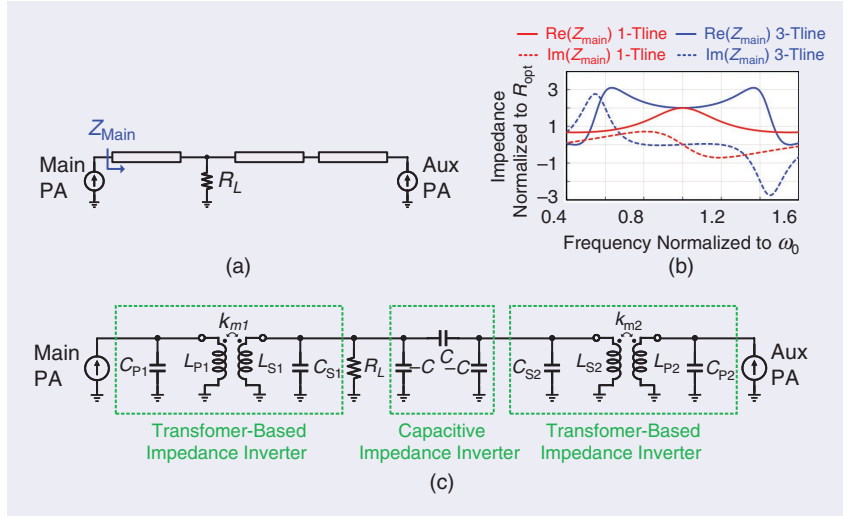


FIGURE 9: (a) Three-Tline Doherty topology, (b) simulated impedance over frequency, and (c) its two-transformer implementation.

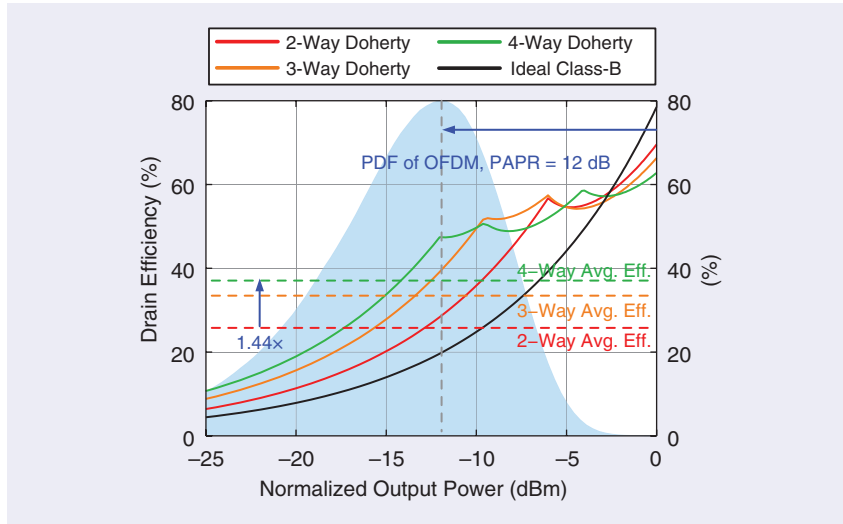


FIGURE 10: Drain efficiency versus normalized output power for two-, three-, and four-way Doherty PAs, overlaid with the PDF of a 12-dB PAPR OFDM signal [12]. The Doherty networks are implemented using Tlines with a practical loss of 0.4 dB/mm, and the transistors are modeled as ideal class-B current sources.

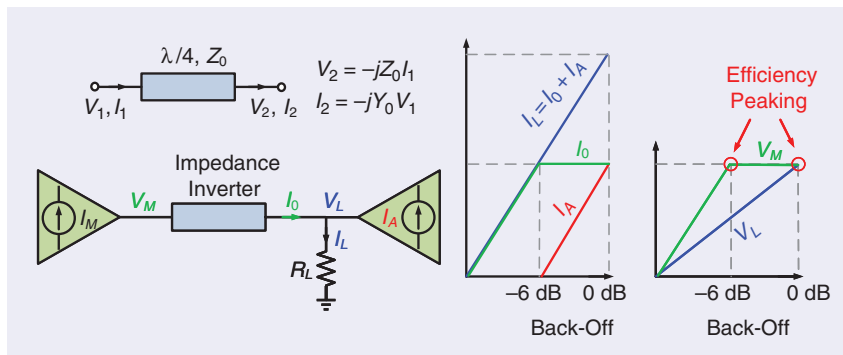


FIGURE 11: Key insight into the $\lambda/4$ Tline.

from the early saturation problem of the cascaded architecture. Meanwhile, the main amplifier's output voltage V_M increases from small signal to 9.5-dB back-off and then saturates, creating the first efficiency peak. Since I_0 follows V_M , the current $I_1 = I_L - I_0$ turns on at 9.5-dB back-off. Through the impedance inverters, V_{A2} and I_{A1} follow I_1 , indicating that 1) auxiliary amplifier #1 turns on at 9.5-dB back-off, and 2) auxiliary amplifier #2 reaches its peak amplitude only at peak power, both agreeing with the desired three-way Doherty operation. Next, I_2 follows V_L , and I_{A2} turns on at 6-dB back-off. As a result, $I_3 = I_2 - I_{A2}$ saturates at 6-dB back-off, which in turn saturates V_{A1} . Up to this point, the voltage and current waveforms of all three amplifiers are derived [see Figure 12(b)], verifying that this architecture indeed achieves the desired three-way Doherty modulation.

A key lesson from comparing the two three-way Doherty architectures is that the cascaded topology treats an existing two-way Doherty as an “aggregated main amplifier” and adds a new auxiliary path, whereas the correct topology treats an existing two-way Doherty as an “aggregated auxiliary amplifier” and adds a new main path. The latter approach makes more sense because it preserves the correct load modulation and efficiency peaks in the back-off region, while the new main path realizes the efficiency peak at peak power.

Inspired by the correct three-way Doherty architecture, we generalize it into a recursive construction rule to build an $(N + 1)$ -way Doherty [2]. Given an existing N -way Doherty, we treat it as an “aggregated auxiliary amplifier” and then combine it with a new main amplifier using an appropriate combiner. The selection of the combiner depends on the output impedance (R_{OUT}) of the N -way aggregated auxiliary. If R_{OUT} presents an open circuit, acting as a current source, the combiner should follow the connections used for the current-source auxiliary in two-way topologies, such as in Figure 13(a), (d), and (e). On the other hand, if R_{OUT} presents a short circuit, acting as a volt-

age source, the combiner should follow the connections used for the voltage-source auxiliary in two-way topologies, as in Figure 13(b) and (c). Applied recursively, from two-way to three-way, three-way to four-way, and so forth, this systematic methodology provides a complete catalog of valid multiway topologies derived from the two-way family [see Figure 13(f)–(j)], with true

Doherty load modulation guaranteed at each step by construction.

A practical barrier to adopting multiway Doherty PAs is the chip area: a conventional N -way Doherty design requires at least N transformers. Recent research has reduced this to $N - 1$ transformers [12], saving one transformer and its associated area and loss. Following this approach,

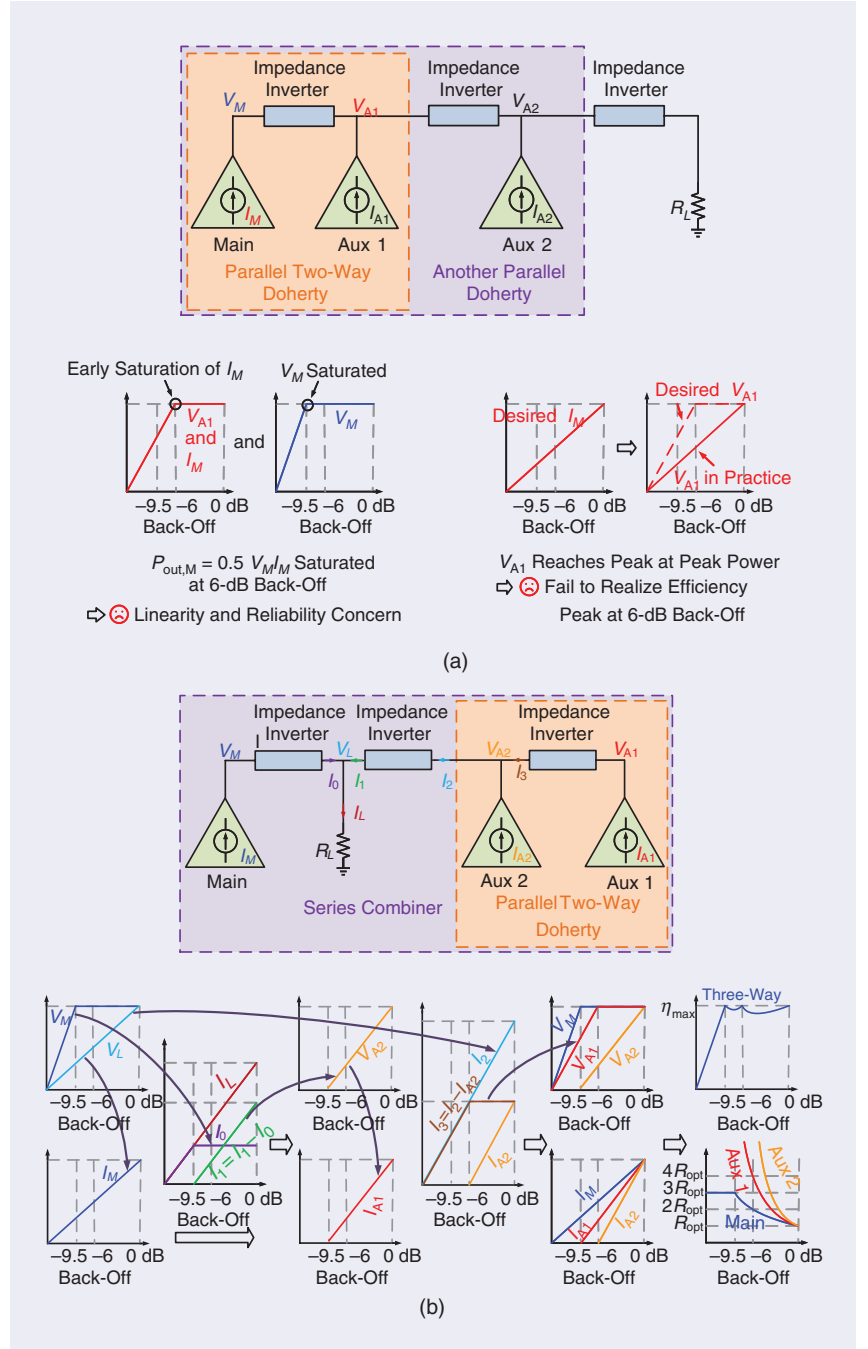


FIGURE 12: (a) The “cascaded” three-way Doherty architecture, which unfortunately cannot realize the desired three-way Doherty load modulation. (b) The correct three-way Doherty architecture [3].

a 47-GHz four-way Doherty PA was demonstrated using only three transformers. The passive loss of the entire four-way Doherty network is only 1.4–1.76 dB across all power levels, which is competitive with a two-stage Wilkinson combiner (typically 1.5–2 dB) that provides four-to-one power combining alone without load modulation. This work also achieved the highest normalized back-off efficiency at 12-dB back-off among reported mmWave PAs at the time of publication [12].

Integrating Doherty PAs Into mmWave Beamformers

As CMOS Doherty PA technology matures, recent research focus is shifting from stand-alone PA demonstrations toward integration within front-end modules (FEMs) [21], [22], [23]. This transition introduces three key design considerations. First, the average efficiency needs to significantly exceed that of class-AB PAs [24] in existing beamformer designs to justify the additional design overhead. Second, since each beamformer consists of multiple identical channels, the size of the Doherty PA needs to be miniaturized, ideally matching that of a class-AB PA. Third, the Doherty PA needs to be seamlessly integrated with the RX LNA and transmit/receive (T/R) switch to ensure an overall high TX efficiency and a low RX NF at the full front-end level.

One effective approach, developed by Samsung [21], [22], adopts the parallel Doherty architecture with an output transformer for $50\text{-}\Omega$ -to- $R_{\text{opt}}/2$ conversion and a CLC impedance inverter (C_Q - L_Q - C_Q in Figure 14). By co-designing the layouts of the transformer and the series inductor L_Q , the entire 28-GHz output network occupies a compact footprint of only $240 \times 240 \text{ }\mu\text{m}^2$. This PA was then integrated into a 32-channel, dual-polarization beamformer IC with a per-channel area comparable to those of prior generations incorporating linear-mode PAs.

In another 28-GHz FEM design [23], the Doherty PA output matching network and the LNA input matching network are co-designed with a T/R switch embedded in the network. The TX fur-

ther employs an asymmetric Doherty architecture with a larger auxiliary amplifier, extending the back-off efficiency enhancement to 9.5 dB. Implemented in the GlobalFoundries 22-nm CMOS SOI process, the FEM occupies a core area of 0.22 mm^2 and delivers 20.3-dBm P_{1dB} and 19.8% PAE at 9.5-dB back-off in the TX mode and 3.1-dB NF in the RX mode.

Conclusion and Future Outlook

The Doherty PA has demonstrated extraordinary longevity. Nine decades after its invention, the two classical architectures (series and parallel Doherty) remain widely adopted, each serving different application scenarios. The successful implementation of a Doherty PA relies on an elegant principle: active load modulation, which increases the load impedance presented to the main amplifier during back-off to sustain the maximum voltage swing. Realizing this in practice requires two key ingredients: a low-loss, compact Doherty output network and cooperation between the main and auxiliary amplifiers. With recent advances in bandwidth extension, multiway Doherty for deep-back-off efficiency enhancement, and beamformer integration, the Doherty PA is extending its reach from cellular infrastructure into new territories.

Looking ahead, the emerging FR3 band (7–24 GHz) offers a favorable tradeoff between coverage and capacity. Optimal architectures, topology choices, semiconductor technologies, and matching network design strategies for this frequency band remain open and impactful research questions. Another challenge is robustness to antenna VSWR variation [25]—a prerequisite for bringing Doherty’s efficiency advantage to user equipment. Importantly, all of these advanced capabilities need to be achieved without incurring a size penalty and without compromising PVT robustness or long-term reliability. This requires size reduction not only of the output network but also of the whole PA chain, including inter-stage matching, drivers, input match-

ing, and I/Q splitting [26]. Finally, with rapid advances in artificial intelligence (AI)-assisted inverse design, discovering how to best leverage AI to improve productivity and optimize performance for Doherty PAs, and thus potentially uncover new PA architectures beyond human heuristics, is an exciting frontier.

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